Jesse McCann

Staff Infrastructure Engineer - Platform, Cloud & AI Infrastructure

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PROFESSIONAL SUMMARY

Staff infrastructure engineer with 15+ years owning production platforms across AWS, GCP, and Azure. I turn manual, fragile operations into one-command automation: multi-cloud Kubernetes, Terraform / OpenTofu infrastructure as code, CI/CD, and cloud cost governance. Recent focus is AI infrastructure and enterprise AI platforms - multi-agent systems, agentic tooling, and the identity and tenancy foundations (standalone GCP tenants, Workforce Identity Federation, MCP workflows) that let teams and AI models ship real code together. Deep FedRAMP High/Moderate and NIST 800-171 experience; mentor to 10+ engineers.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS, GCP, Azure, Kubernetes (GKE), Terraform / OpenTofu, Ansible, Docker, Linux

Languages: Python, Go, Bash, PowerShell, SQL

Identity & Access: Microsoft Entra ID, Workforce Identity Federation, OIDC, SAML, least-privilege IAM, SCIM

AI Platform & Infrastructure: Gemini Enterprise (Agentspace), Vertex AI, GitHub Copilot, multi-agent orchestration, MCP / agentic tooling

Automation & Integration: Workato, Workday, CI/CD (GitHub Actions, Cloud Build, GitLab CI), mise, uvx, Git

Observability & Data: OpenSearch, Apache Kafka, ELK Stack, PRTG

Security & Compliance: FedRAMP (High/Moderate), NIST 800-171, Zero Trust, vulnerability management, continuous monitoring

PROFESSIONAL EXPERIENCE

Senior Systems Infrastructure Engineer, Corporate IT (Consultant)

BlackLine - Greater Philadelphia | 2025 - Present

Consulting at the intersection of Corporate IT and AI engineering; building BlackLine IT's enterprise AI platform foundation.

- Launched BlackLine's Gemini Enterprise (Agentspace) platform: stood up Dev, Test, and Prod environments and data connectors (SharePoint, OneDrive, Outlook, Teams, Jira, Confluence), moved apps from SAML to OIDC, and restricted production data access.
- Designed and delivered a standalone, Terraform-managed GCP tenant for AI and agent workloads - org and folder structure, least-privilege IAM for citizen developers, network and tenant isolation, and a reusable Terraform module for business-unit agent projects.
- Implemented Workforce Identity Federation between Microsoft Entra ID and GCP; added a vaulted break-glass account and dedicated elevated accounts, and removed over-privileged ("god-mode") bindings and legacy admin tiers.
- Established infrastructure-as-code and release discipline - migrated Terraform state, centralized shared configuration, consolidated Entra identity management, and added semantic versioning, Conventional Commits, automated changelogs, and release pipelines; built self-service AI Playgrounds with on-demand provisioning, automatic TTL cleanup, and a v1.0.0 library of reusable AI skills.
- Automated the employee lifecycle (PowerShell onboarding/offboarding, scheduled Workday export, daily reports, PRTG monitoring); delivered Workato SSO/SCIM and a Confluence OAuth connector; authored ~100 Confluence pages of architecture and runbooks.

Site Reliability Engineer, DAS Cloud

DTEX Systems - Greater Philadelphia | May 2024 - Present

Reliability and platform engineering for multi-tenant DAS Cloud across commercial and FedRAMP High/Moderate environments.

- Delivered DAS Cloud reliability and platform work across ~300 completed Jira issues (SRE, Service Desk, DAS 7, FedRAMP, customer environments); authored ~126 Confluence pages of architecture, runbooks, and operational standards.
- Built a multi-cloud Kubernetes deployment platform (Terraform / OpenTofu) that cut customer-environment provisioning from days of manual work to one command and ~1 hour, and reduced monthly cloud spend 45% through Cloud Custodian policy automation.
- Designed an internal multi-agent AI platform (Vertex AI + Copilot) with a durable issue graph for long-running agent work, now the team's blueprint for large-scale legacy refactoring.
- Defined Terraform style and module standards, introduced a facade-based infrastructure pattern, split the SRE monorepo into independently managed repositories, and shipped the DAS 7.3 Terraform and Ansible release.
- Closed critical FedRAMP controls (SSH hardening, key rotation, SAML protection for Nessus, scanner evidence, Salt/Zabbix remediation, GitLab upgrades) and expanded production IAM controls across RDS, CloudWatch, logging, Lambda, and snapshots.
- Evaluated and delivered multiple OpenSearch architectures (AWS, Aiven, Canonical Charmed with Juju), adopting Cluster Insights, UltraWarm/cold storage, and in-place DAS upgrades across customer, QE, POV, and internal environments; completed GCP commercial productionization and documented repeatable DAS 7 deployment, image-pipeline, and SSH-over-IAP patterns.
- Improved detection and recovery via DAS CloudWatch alerting, CloudWatch-to-Slack integration, a service-status page, RDS snapshots, immutable backups, and multi-AZ conversion; led resolution of customer incidents involving Elasticsearch health, storage, certificate expiry, and message-queue pressure.

Sr. Cloud Systems Engineer / Technical Lead

Confluent - Mountain View, CA | Nov 2020 - Apr 2024

Infrastructure through pre-IPO, IPO, and post-IPO growth at the company behind enterprise Apache Kafka.

- Built and operated Go and Python infrastructure services on GCP Cloud Run, with CI/CD from GitHub through Cloud Build; deployed GKE with Terraform and custom Go cluster-management tooling.
- Led a multi-quarter Zero Trust network rollout (Palo Alto) for a global workforce, writing the allow-list automation so the cutover caused zero work interruptions.
- Killed a proposed GitHub EMU migration by building proof-of-concept tooling to quantify its blast radius; presented findings to InfoSec, TPM, and Engineering leadership to steer the decision to GitHub Enterprise. Mentored 10 engineers across the infrastructure teams.

Staff System Engineer

BlackLine - Woodland Hills, CA | Feb 2018 - Nov 2020

Migrated 4 Nutanix clusters, 30 servers, and 10 applications to GCP with Terraform; ran the Atlassian stack (3,000 users) in high availability and automated HRIS updates, server deployment, and patch management, saving 10-15 hours per week.

Earlier Roles

Senior System Engineer, Applied Minds (2014-2018) - NIST 800-171 compliance and AWS migration at 99.99% uptime, with real-time ELK monitoring; promoted to IT Manager. **Senior Systems Engineer, Virgin Galactic** (2012-2014) - high-availability clusters and network security. **Server Support Engineer, Publix Super Markets** (2007-2012).

EDUCATION, CERTIFICATIONS & PUBLICATIONS

AWS Certified Solutions Architect - Associate | Technical Mentor, Princeton University | Polk State College, 2005-2006 | CFCC, 1999-2001